

HOW SECURE IS YOUR AZURE DATABRICKS? LET'S FIND OUT

Fabian Gutierrez Ards
Data Saturday Holland 2025



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Fabian Gutierrez Ardt

Consultant | Azure & Databricks Data Engineer | Freelancer



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consultancy

20

20 years in the IT world, almost 15 years in Data world



Data Communities, Outdoor activities & travelling



linkedin.com/in/fabiangutierrezardts/



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WHY TALK ABOUT DATABRICKS SECURITY?

Working with my costumers, I came across several challenges trying to introduce or implement (Azure) Databricks when getting approval from C/ISO (Chief / Information Security Officer) and Architects.

Databricks and Microsoft provided me with valuable information and a powerful Tool to overcome this challenges so that I could collect and present evidence and gain approvals.

Free your Data



TOPICS

- SAT Background
- Let's install SAT
- Evaluate and stay up to date
- Extra Security Measurements

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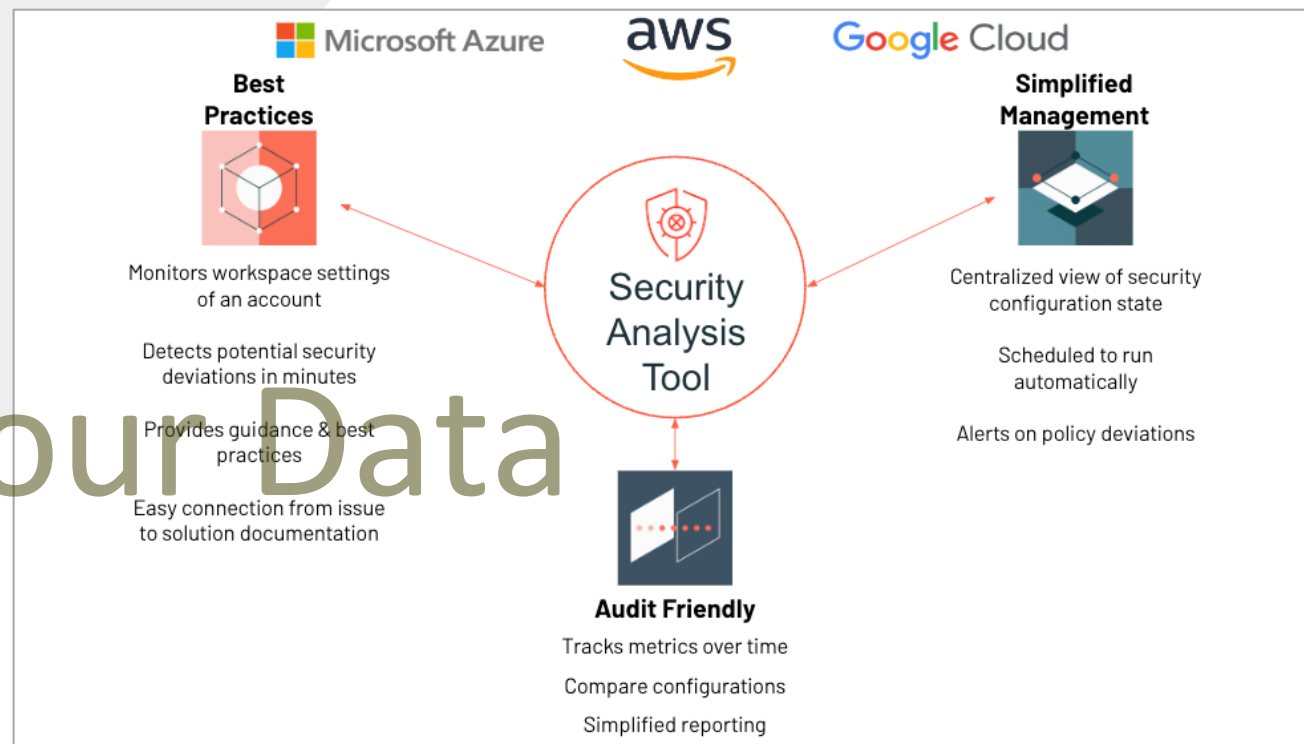
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SAT BACKGROUND

WHAT IS SAT?

It stands for **Security Analysis Tool**

SAT checks for typical security configurations deployed by most high-security organizations. It highlights the most significant risks, and those most frequently asked about by customers. Each recommendation will include a security configuration reference link to the relevant Databricks documentation.



Is an observability utility designed to improve the security posture of Databricks deployments. It helps customers identify deviations from established security best practices and monitor the security health of their Databricks workspaces.

When to use SAT?

COMMON SCENARIOS ARE:

- YOUR COMPANY REQUIRES CONSTANT SECURITY MONITORING.
- HAVE INDUSTRY SECURITY BEST PRACTICES APPLIED.
- EVALUATE FROM ACCOUNT OR INDIVIDUAL WORKSPACES LEVEL.
- PROVIDE KEY ROLE SECURITY FIGURES WITH TANGIBLE EVIDENCE.

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SAT IS NOT FORMALLY SUPPORTED

“SAT is a productivity tool designed to help verify your security configurations against Databricks’ best practices. It is *not* intended to serve as a certification or formal attestation of your deployment's security posture.”

<https://databricks-industry-solutions.github.io/security-analysis-tool/docs/motivation/>

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The code in this project is provided **for exploration purposes only** and is **not formally supported** by Databricks under any Service Level Agreements (SLAs). It is provided **AS-IS**, without any warranties or guarantees.

WHAT IS BEING MONITORED?

- Network Security (NS)



- Governance (GOV)



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- Data Protection (DP)



- Identity & Access (IA)



- Informational (INFO)



AND...EXTRA SELF ASSESMENT CHECKS

- Object storage encryption for data sources

- Enable single sign-on

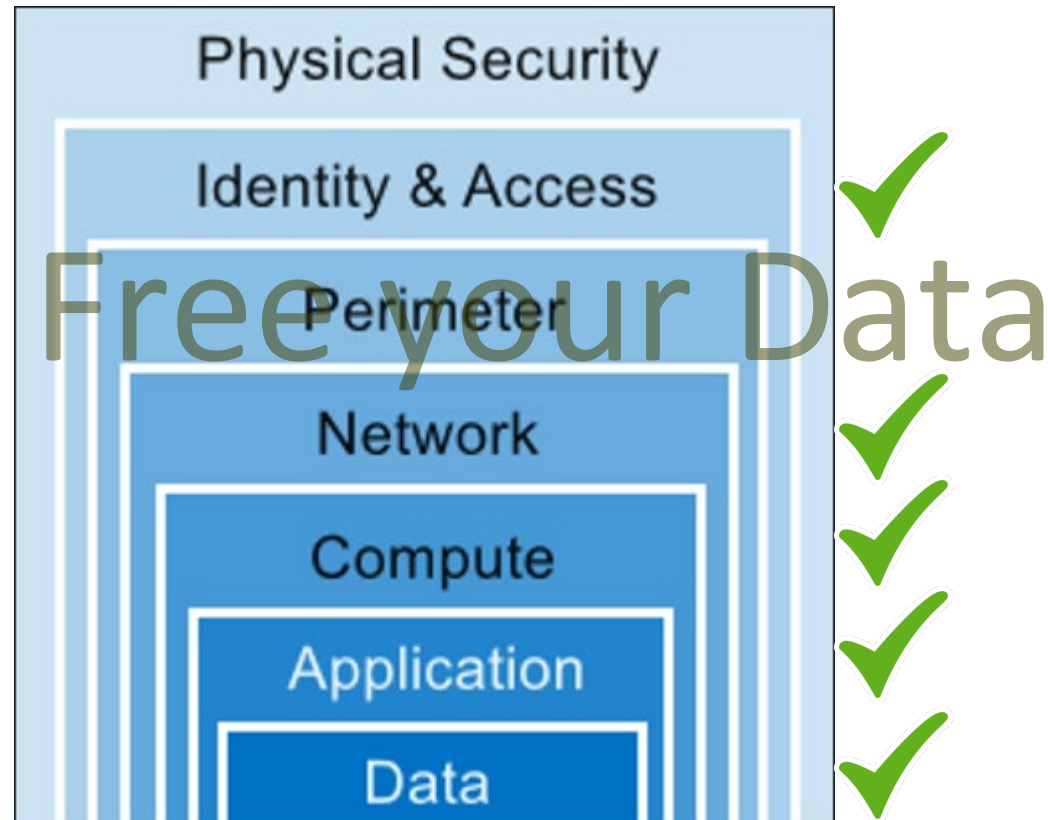
Free your Data

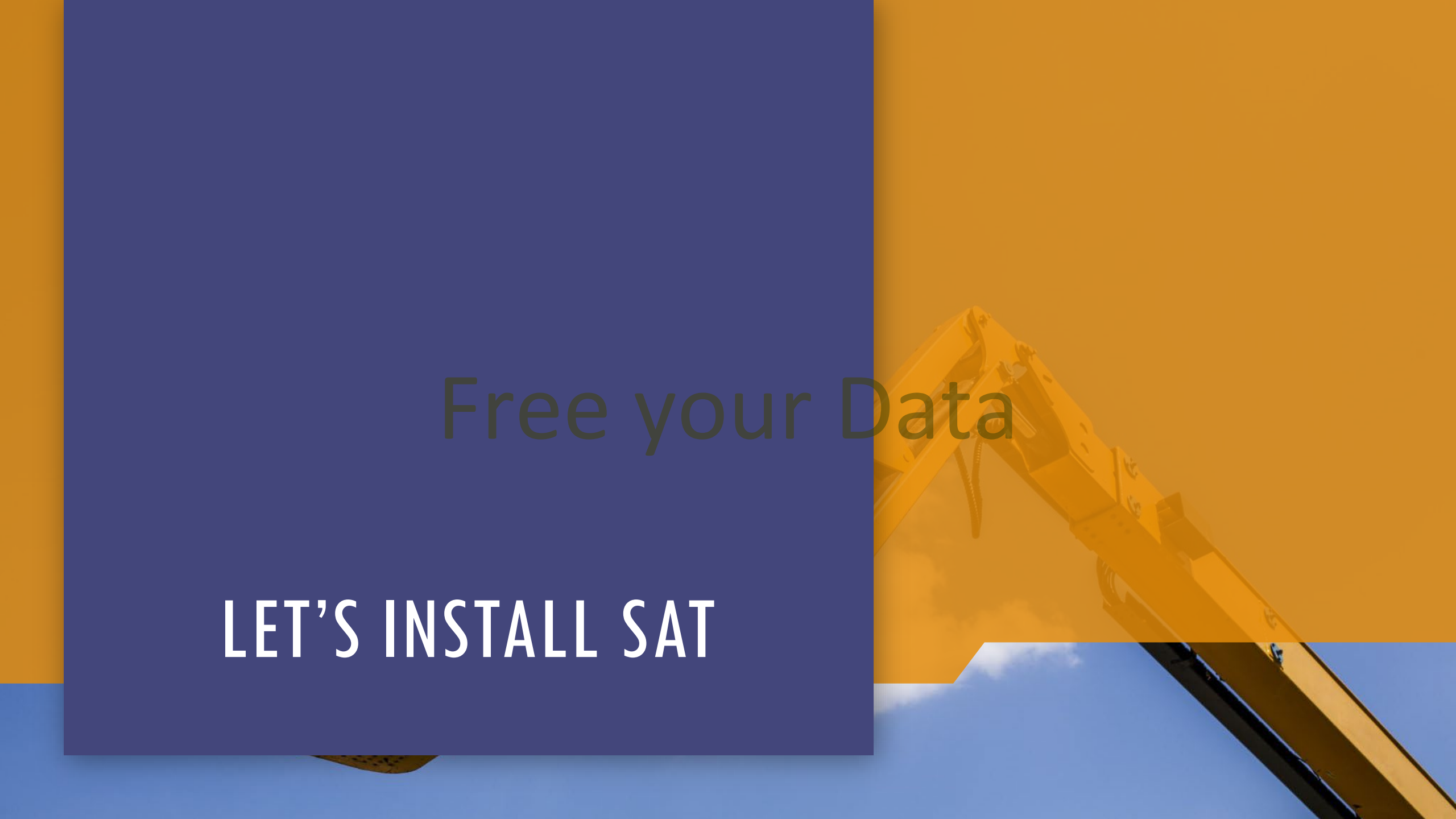
- SCIM (System for Cross-Domain Identity Management) for user provisioning

- Table Access Control for clusters that don't use Unity Catalog

- Detect network peering of VPC (AWS, GCP) or VNet (Azure)

COMPARING WITH DEFENSE IN DEPTH





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LET'S INSTALL SAT

REQUIREMENTS

- An Entra App Registry to be used as Identity with privilege permission on
 - Azure Subscription Owner (RBAC)
 - Databricks Account
 - All Databricks Workspaces where analysis required.

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- Python 3.9 or higher
- Databricks CLI installed with a profile logged.
- Databricks SQL Warehouse (To run the SQL dashboard)
- PyPI access from your workspace (To install the SAT utility library)
 - <https://pypi.org/project/dbl-sat-sdk/>
 - You can also modify SAT code to handle isolated installation if your workspace has no Public access

EXTRA REQUIREMENTS

- Databricks Configuration file (.dabrickscfg).
- Not mandatory but good advise, have a dedicated **monitoring** workspace to host the solution.
 - This will help you to have centralized SAT solution.
 - Compute within isolated workspace.
- GitHub company (preferences) account or individual account.
 - This will be very helpful when questions or possibles issues.
- Knowledge handling DABS (Databricks Asset Bundles)
 - Micro Bundles

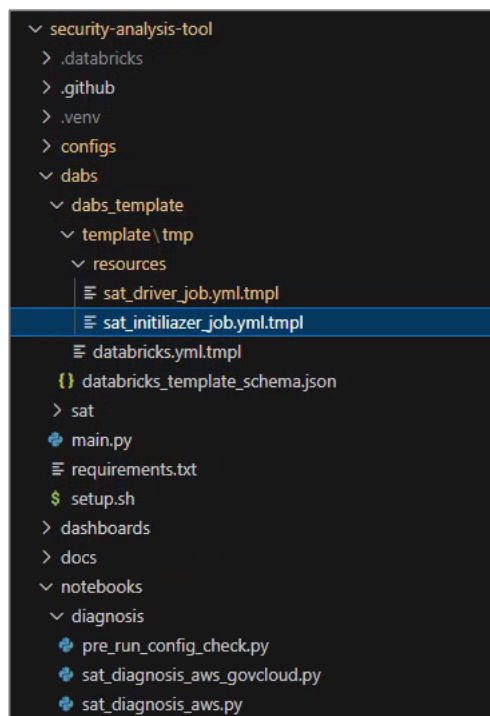


BEFORE INSTALLING SAT

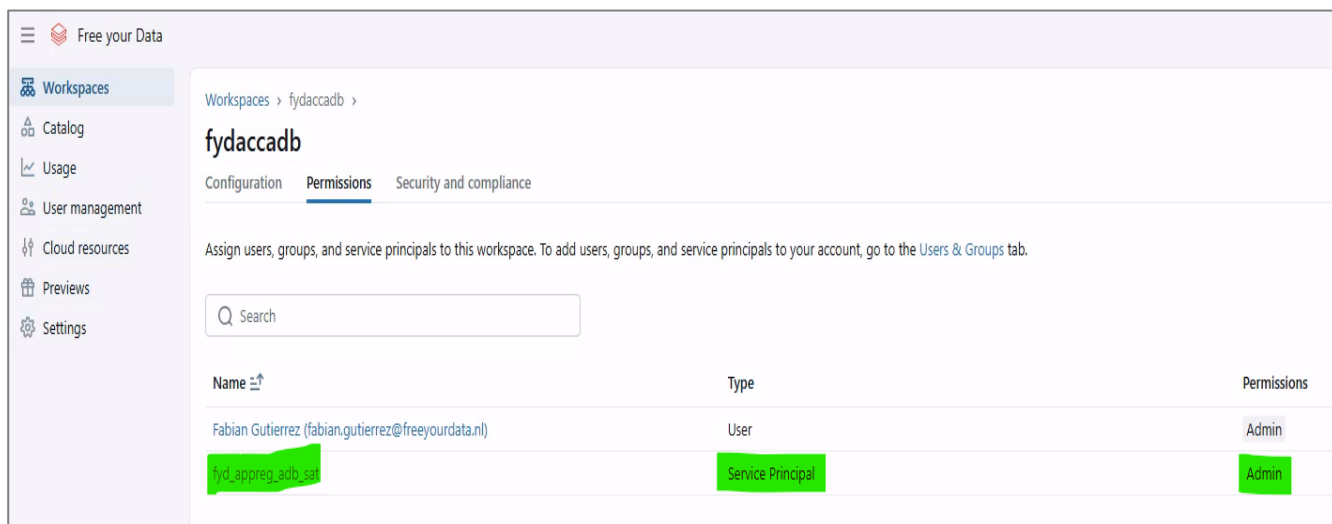
- Study the Job YAML and adjust the Workers type that suits your region and Compute usage.

Adjust job cluster via:

Source Code \ dabs \ dabs_template \ template \ tmp \ resources \ sat_initializer_job.yml.tmpl
 \ sat_driver_job.yml.tmpl

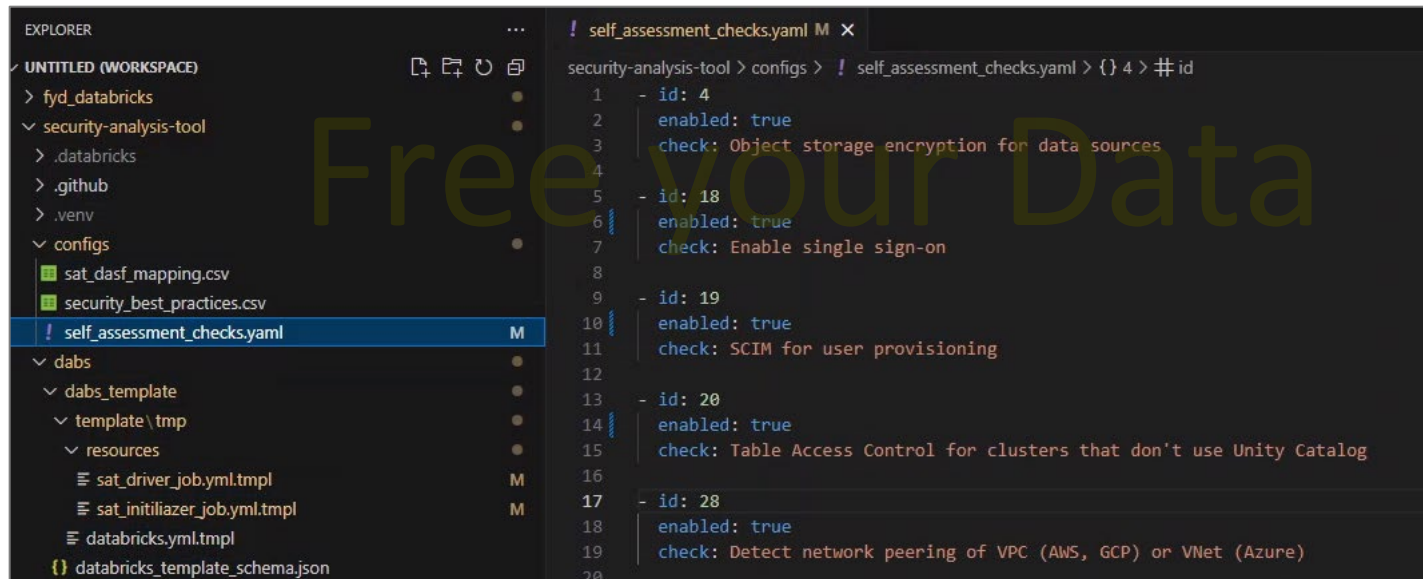


- Provide Workspace admin access to Service account to all workspace where analysis required.



BEFORE INSTALLING SAT

If you are familiar with the (manual) self assessments checks, then you can enable them before installing or after. It depends on what you want to evaluate.

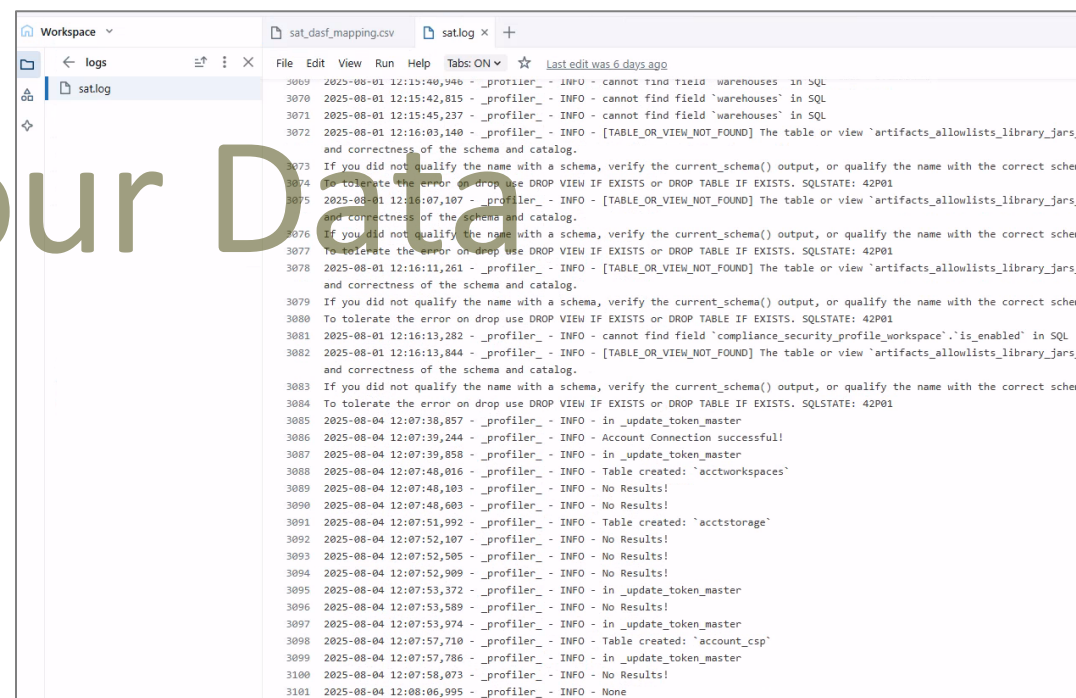
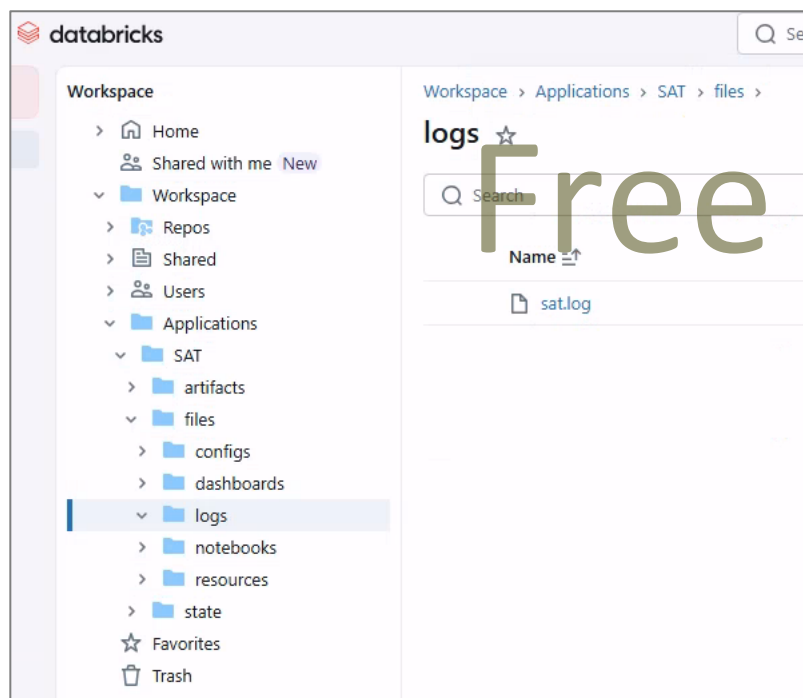


```
EXPLORER
UNTITLED (WORKSPACE)
  > fyd_databricks
  > security-analysis-tool
    > .databricks
    > .github
    > .venv
    > configs
      sat_dasf_mapping.csv
      security_best_practices.csv
      ! self_assessment_checks.yaml M
    > dabs
      > dabs_template
        > template\
          > resources
            sat_driver_job.yml.tpl M
            sat_initializer_job.yml.tpl M
            databricks.yml.tpl
            {} databricks_template_schema.json

! self_assessment_checks.yaml M X
security-analysis-tool > configs > ! self_assessment_checks.yaml > {} 4 > # id
1 - id: 4
2   enabled: true
3   check: Object storage encryption for data sources
4
5 - id: 18
6   enabled: true
7   check: Enable single sign-on
8
9 - id: 19
10  enabled: true
11  check: SCIM for user provisioning
12
13 - id: 20
14  enabled: true
15  check: Table Access Control for clusters that don't use Unity Catalog
16
17 - id: 28
18  enabled: true
19  check: Detect network peering of VPC (AWS, GCP) or VNet (Azure)
20
```

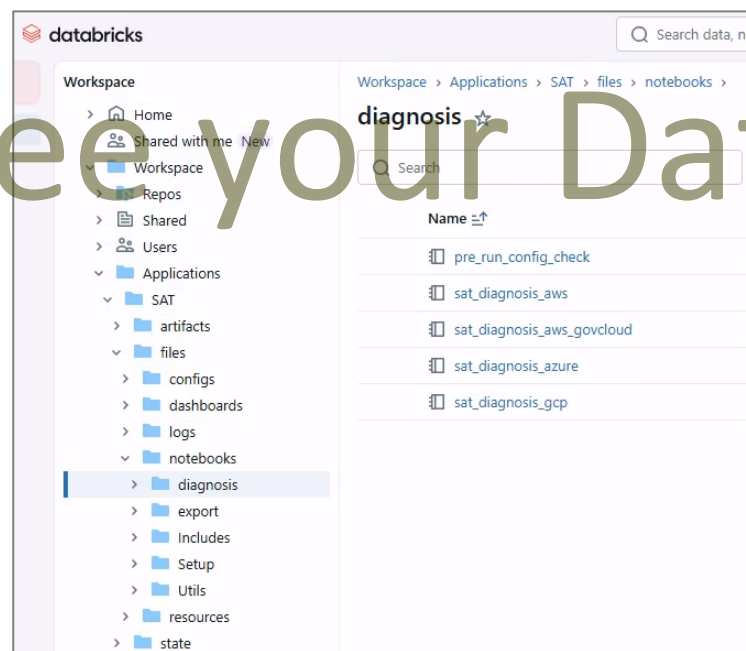
DURING INSTALLATION

- Logging (Workspace -> logs)



DURING INSTALLATION

- Make use of Diagnosis Notebooks to troubleshoot during installation.
Very helpful when fixing installation/configuration issues.



AFTER INSTALLING SAT

Run Jobs:

- Initial Job [Manual Execution & 1 time only]
- SAT Driver [3 days a week Mo-Wed-Fr]

Jobs & Pipelines

▼ Create new

 **Ingestion pipeline**
Ingest data from popular apps, databases and file sources

 **ETL pipeline**
Build ETL pipelines using SQL and Python

Jobs & pipelines Job runs

🔍 Filter by name or ID substring

All Jobs Pipelines Owned by me Accepted

Name ↕	Type	Tags
SAT Driver Notebook	Job	SAT
SAT Initializer Notebook (one-time)	Job	SAT

Jobs & Pipelines > SAT Initializer Notebook (one-time) > Run 507483981268927 >

sat_initializer run ✔ Succeeded Lakeflow Jobs UI: OFF ▼

Output

✓ 4m 58s 9

```
notebooks = [  
  ("1. list_account_workspaces_to_conf_file", 3000),  
  ("3. test_connections", 12000),  
  ("4. enable_workspaces_for_sat", 3000),  
  ("5. import_dashboard_template_lakeview", 3000),  
  ("6. configure_alerts_template", 3000),  
  ("9. self_assess_workspace_configuration", 3000),  
]  
  
for notebook, timeout in notebooks:  
  status=run_notebook(F"{basePath()}/notebooks/Setup/{notebook}", timeout)
```

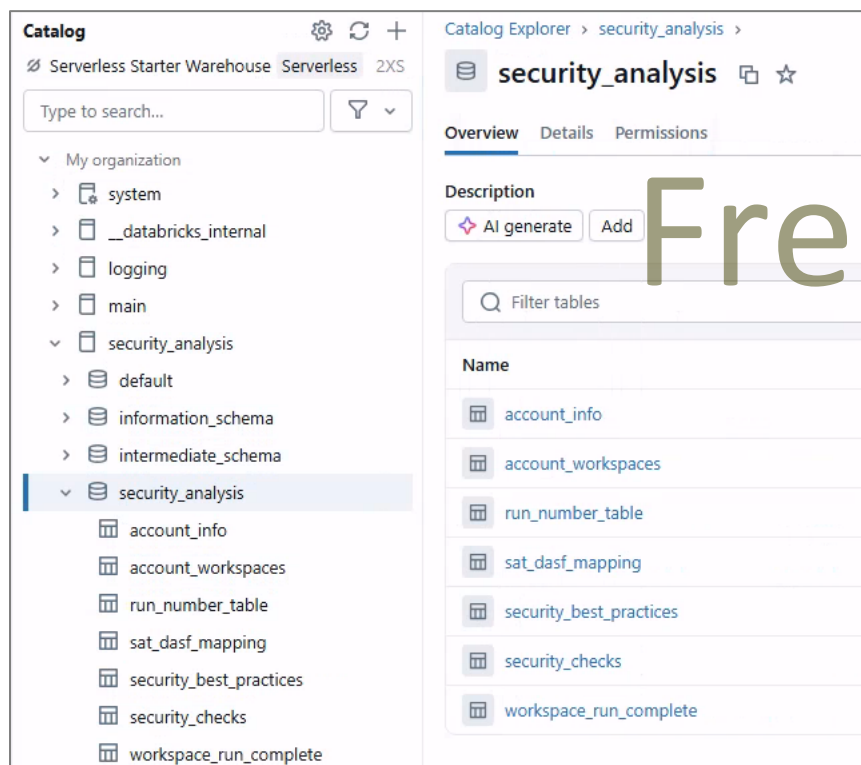
Notebook Workflows

Notebook path	Status
/Workspace/Applications/SAT/files/notebooks/Setup/1. list_account_workspaces_to_conf_file	1 ✔ Succeeded
/Workspace/Applications/SAT/files/notebooks/Setup/3. test_connections	3 ✔ Succeeded
/Workspace/Applications/SAT/files/notebooks/Setup/4. enable_workspaces_for_sat	2 ✔ Succeeded
/Workspace/Applications/SAT/files/notebooks/Setup/5. import_dashboard_template_lakeview	3 ✔ Succeeded
/Workspace/Applications/SAT/files/notebooks/Setup/6. configure_alerts_template	3 ✔ Succeeded
/Workspace/Applications/SAT/files/notebooks/Setup/9. self_assess_workspace_configuration	3 ✔ Succeeded

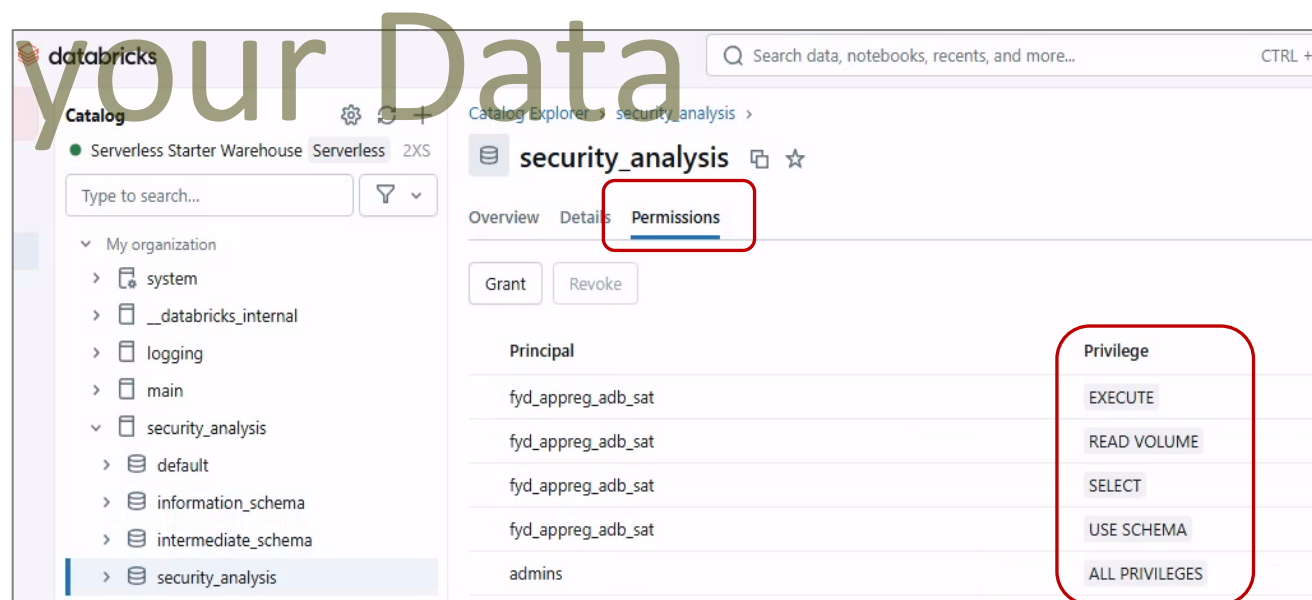
In case of
errors, use
again Diagnosis
Notebooks

AFTER INSTALLING SAT

Explorer Catalog: Where all this information stored



- Ensure the users have the necessary privileges on Schema level to ensure they can read data.





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DEMO TIME

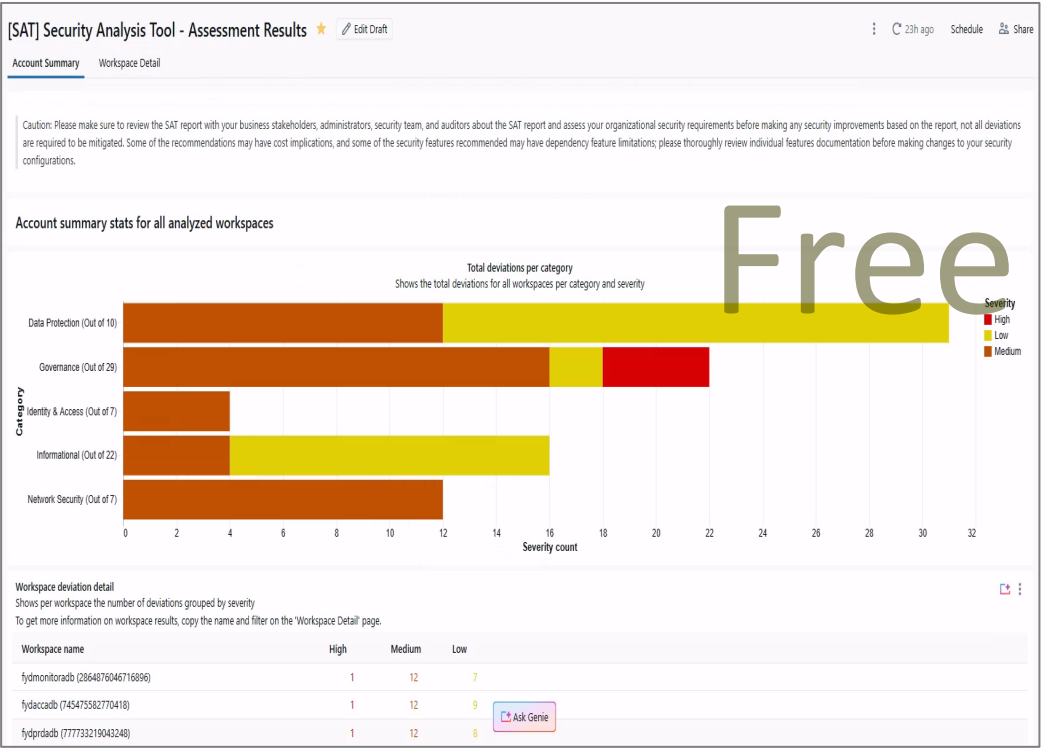


Free your Data

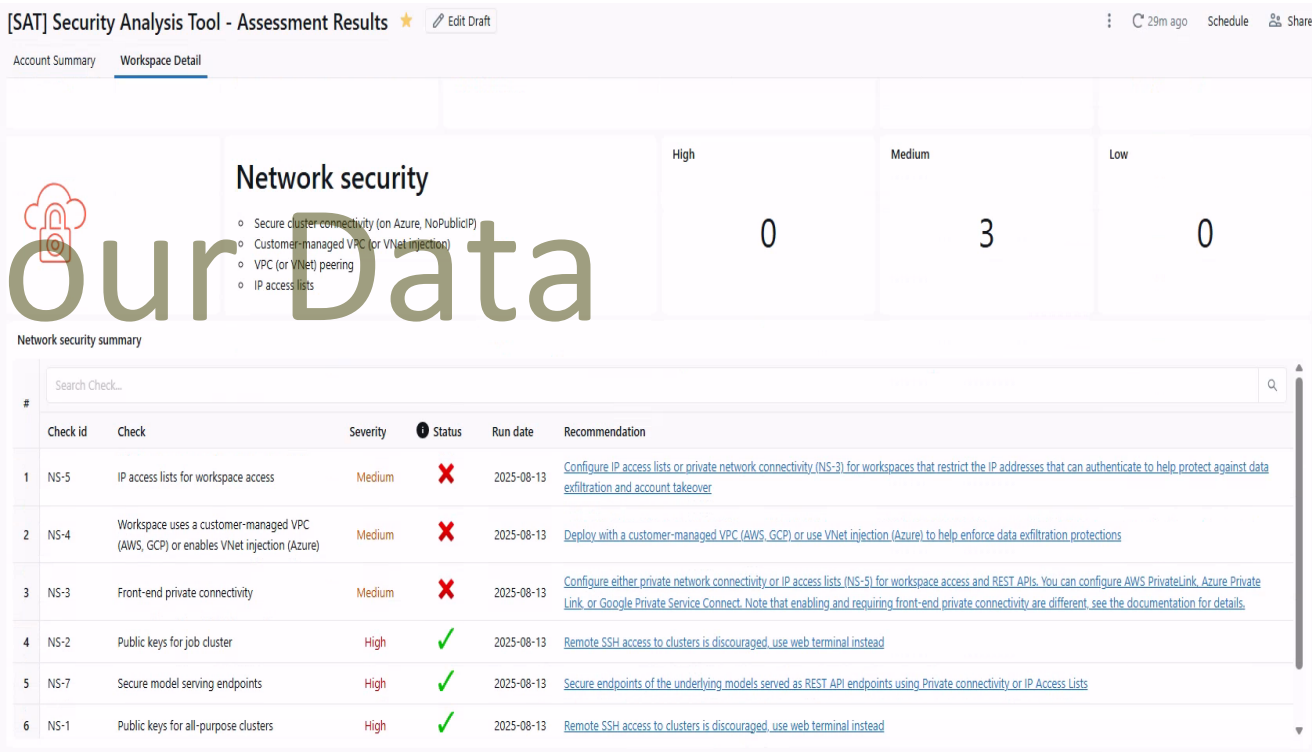
**EVALUATE AND STAY
UP TO DATE**

ASSEST SECURITY ANALYSIS

Account summary



Per Workspace



APPLYING SECURITY MEASUREMENTS

[SAT] Security Analysis Tool - Assessment Results ★ Edit Draft 44m ago Schedule Sha

Account Summary **Workspace Detail**

7 NS-6 [Secure cluster connectivity \(on Azure this is also called NoPublicIp\)](#) Medium ✓ 2025-08-06 [Configure secure cluster connectivity \(On Azure, this is called NoPublicIP / NPIP\)](#)

 **Identity & Access**

- Single sign-on (AWS only, Azure and GCP always enabled)
- SCIM for user provisioning
- Role-based access control
- Token management

High 0 Medium 1 Low 0

Identity & Access summary

Search Check...

#	Check id	Check	Severity	Status	Run date	Recommendation
1	IA-5	Maximum lifetime of new tokens to something other than unlimited	Medium	✗	2025-08-06	Configure maximum lifetime for all future tokens to a value other than unlimited
2	IA-2	SCIM for user provisioning	High	✓	2025-08-06	Provision users from your Identity Provider with workspace-level SCIM APIs
3	IA-3	Table Access Control for clusters that don't use Unity Catalog	High	✓	2025-08-06	For clusters that do not use Unity Catalog, enable Table Access Control in admin settings so that you can utilize Table ACL clusters that enforce user isolation
4	IA-1	Enable single sign-on	High	✓	2025-08-06	Authenticate via single sign-on and leverage multi-factor authentication
5	IA-4	PAT tokens with no lifetime (expiration) limit	Medium	✓	2025-08-06	Configure maximum token lifetimes for future tokens using token management



Each check is provided with an recommendation and a link to the solution page. Follow the instructions, apply changes where required and on this next job run the status will be updated.

ADDITIONAL DETAILS

Use this to get details on what caused a check to fail.

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[SAT] Security Analysis Tool - Assessment Results

34m agoScheduleShare

Exit fullscreen

Additional details
Allows for searching information of a specific Security Check. Use search bar for filtering

Search Check ID...

Check ID	Check	Recommendation	Additional Details	Logic	Evaluation Value	Api
IA-7	Use service principals	Run production workloads with service principals	{ ... } // 1 item	Check if at least one service principal exists for the ...	1	curl --netrc -X GET https://adb-229099765822
DP-1	Secrets management	Store and use secrets securely	{ } // 0 items	Check if there are any secrets configured in the wor...	1	curl --netrc -X GET https://adb-229099765822
GOV-10	Managed tables in DBFS root	The DBFS root is not intended for production customer d...	{ ... } // 1 item	Check the /user/hive/warehouse/ folder for any data...	1	curl --netrc -X GET https://adb-229099765822
GOV-11	DBFS mounts	Avoid DBFS mounts for accessing production data	{ ... } // 1 item	Check for mnt paths in dbutils.fs.mounts() and repor...	1	dbutils.fs.mounts()
INFO-6	Number of admins	Limit the number of admin accounts so that most users a...	{ ... } // 1 item	Check members count of displayName "admins" in g...	2	curl --netrc -X GET https://adb-229099765822
GOV-4	Long-running clusters	Restart clusters on a regular schedule to use the latest av...	{ ... } // 1 item	Check each running cluster's last restart time till no...	24	curl --netrc -X GET https://adb-229099765822
INFO-2	Set maximum concurrent runs in job c...	Limit the number of parallel runs for a given job to avoid ...	{ ... } // 1 item	Check if max_concurrent_runs configuration for eac...	5	curl --netrc -X GET https://adb-229099765822
INFO-4	Number of users who have cluster cre...	Limit number of users with cluster create privileges. Note...	{ ... } // 1 item	Check entitlements.value for each user and look if th...	5	curl --netrc -X GET https://adb-229099765822
GOV-2	PAT tokens are about to expire	Set an expiration day (maximum lifetime) for tokens. Also...	{ ... } // 1 item	Check each token expiry_time and report if the expir...	7	curl --netrc -X GET https://adb-229099765822
IA-4	PAT tokens with no lifetime expiration	Configure maximum token lifetimes for future tokens usi...	{ ... } // 1 item	Check each token expiry_time and report if the expir...	90	curl --netrc -X GET https://adb-229099765822
NS-1	Public keys for all-purpose clusters	Remote SSH access to clusters is discouraged, use web te...	{ ... } // 1 item	Check if ssh_public_keys is configured on any cluster	N/A	curl --netrc -X GET https://adb-229099765822
NS-2	Public keys for job cluster	Remote SSH access to clusters is discouraged, use web te...	{ ... } // 1 item	Check if ssh_public_keys is configured on any job cl...	N/A	curl --netrc -X GET https://adb-229099765822
NS-3	Front-end private connectivity	Configure either private network connectivity or IP access...	{ ... } // 1 item	Check if private_access_settings_id is set for the wor...	N/A	curl -n -X GET https://accounts.cloud.databricks
NS-4	Workspace uses a customer-managed ...	Deploy with a customer-managed VPC (AWS, GCP) or use...	{ ... } // 1 item	Check if network_id is set for this workspace	N/A	curl -n -X GET https://accounts.cloud.databrick
NS-5	IP access lists for workspace access	Configure IP access lists or private network connectivity (...)	{ ... } // 1 item	Check if ip-access-lists are configured and enabled	N/A	curl --netrc -X GET https://adb-229099765822
NS-6	Secure cluster connectivity (on Azure t...	Configure secure cluster connectivity (On Azure, this is ca...	{ ... } // 1 item	Check if enableNoPublicIp are configured and enabl...	N/A	curl --netrc -X GET https://accounts.azuredata
INFO-7	Detect network peering of VPC (AWS, ...)	Network peering is discouraged because private endpoint...	{ ... } // 1 item	Manual check (Update configuration status using <SA...	N/A	Manual check

[SAT] Security Analysis Tool - Assessment Results

41m agoScheduleShare

Exit fullscreen

Additional details
Allows for searching information of a specific Security Check. Use search bar for filtering

NS-4

Check ID	Check	Recommendation	Additional Details	Logic	E	Api
NS-4	Workspace u...	Deploy with a customer-managed VPC (AWS, ...	{ "workspaceId": "2290997658222571" }	Check if network_id is set for this workspace	N..	curl -n -X GET 'https://accounts.cloud.databricks.com/api/2.0/accounts/<account_id>/workspace

COMPARE EVALUATIONS BY DATES

Its possible to compare 2 different run dates to track progress or deterioration of Workspaces as changes get applied or omitted.

[SAT] Security Analysis Tool - Assessment Results ★ [Edit Draft](#) 1m ago [Schedule](#) [Share](#)

[Account Summary](#) [Workspace Detail](#)

Previous run date
Select the date of another run to compare scores

14 days ago (Jul 30, 2025)

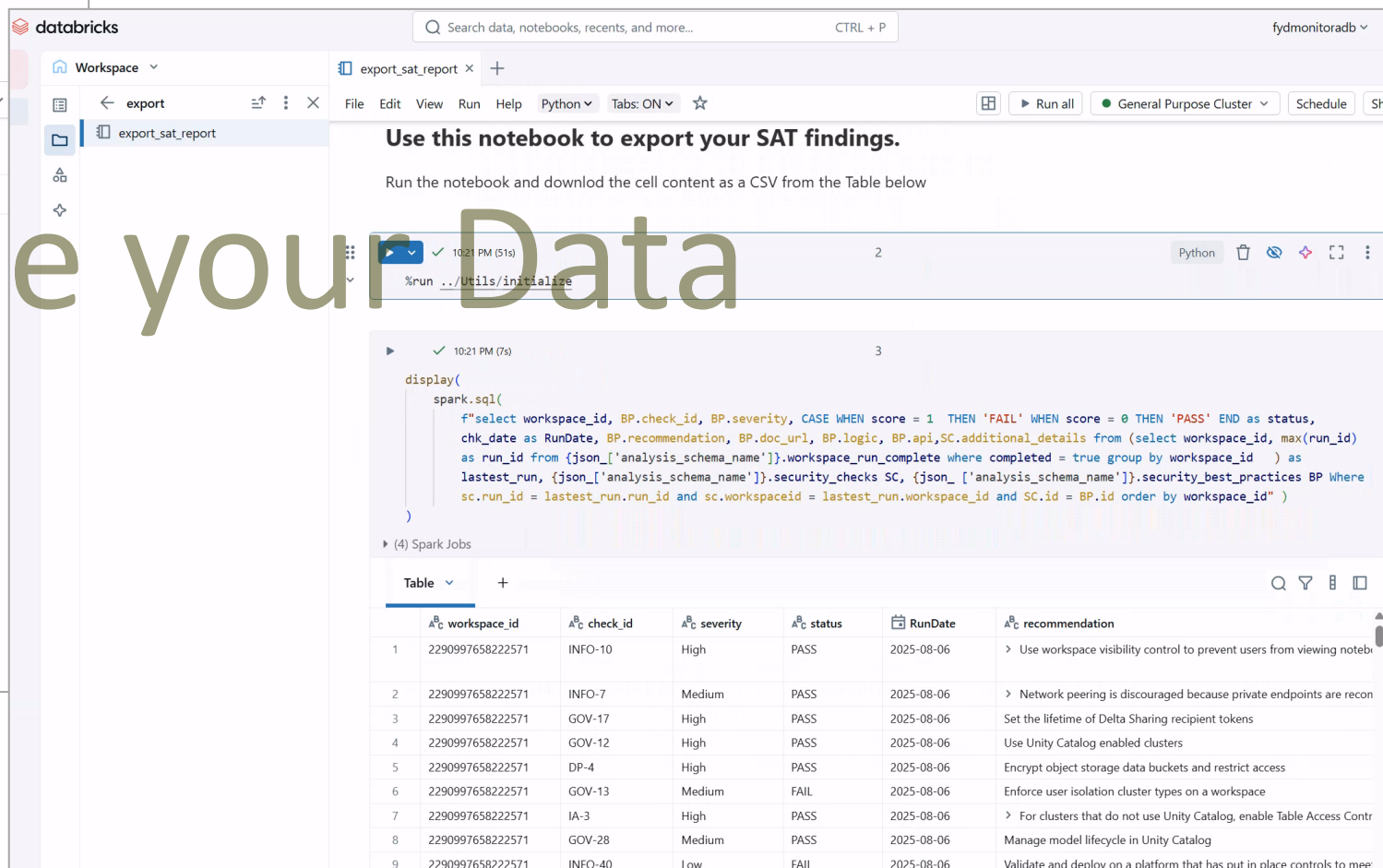
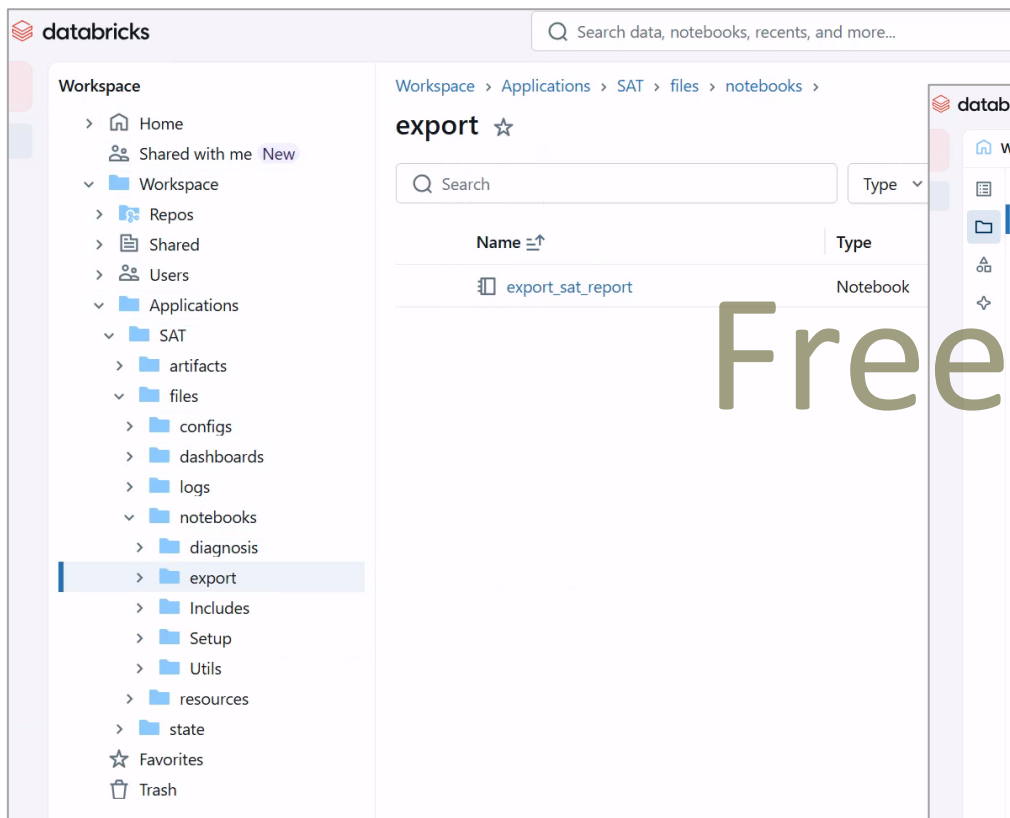
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Security configuration comparison

#	Check id	Check	Severity	Run 1 stati	Run 2 stati	Run1 date	Run 2 date	Recommendation
51	INFO-11	Workspace for supporting Git repos	High	✓	✓	2025-08-13 21:56:57	2025-07-30 12:17:42	It is recommended to store code in Git repos
52	DP-5	Downloading results is disabled	Medium	✓	✗	2025-08-13 21:57:02	2025-07-30 12:17:46	Disable download button for notebook results
53	IA-5	Maximum lifetime of new tokens to something other than unlimited	Medium	✗	✗	2025-08-13 21:57:05	2025-07-30 12:17:51	Configure maximum lifetime for all future token
54	GOV-13	Enforce User Isolation	Medium	✗	✗	2025-08-13 21:57:09	2025-07-30 12:17:56	Enforce user isolation cluster types on a worksp
55	DP-6	Notebook export	Low	✗	✗	2025-08-13 21:57:11	2025-07-30 12:18:00	Disable exporting notebooks and cells within n
56	DP-7	Notebook table clipboard features	Low	✓	✗	2025-08-13 21:57:13	2025-07-30 12:18:02	Disable the ability of users to copy tabular data
57	DP-8	Enable storing interactive notebook results only in the customer account	Medium	✗	✗	2025-08-13 21:57:17	2025-07-30 12:18:07	Enable store interactive notebook results in you
58	DP-9	FileStore endpoint for HTTPS file serving	Medium	✗	✗	2025-08-13 21:57:23	2025-07-30 12:18:13	Review and disable FileStore endpoint in admin
59	GOV-15	Enable verbose audit logs (on Azure, diagnostic logs)	Medium	✗	✗	2025-08-13 21:57:20	2025-07-30 12:18:09	Enable verbose audit logs (on Azure, diagnostic
60	GOV-24	Legacy global init script	High	✓	✓	2025-08-13 21:57:25	2025-07-30 12:18:15	Databricks recommends that you migrate legac
61	GOV-26	Legacy cluster-named init scripts	High	✓	✓	2025-08-13 21:57:27	2025-07-30 12:18:18	Databricks recommends that you migrate legac

EXPORTS

- Export all Workspaces, unless you modify the query.
- Status based on ≥ 1 within the query.



ALERTS

Name	Status	Last updated
sat_alerts_777733219043248	TRIGGERED	1 minute ago
sat_alerts_745475582770418	TRIGGERED	1 minute ago
sat_alerts_2290997658222571	TRIGGERED	59 seconds ago
sat_alerts_2864876046716896	OK	32 seconds ago

- 1 alert per Workspace.
- Recommendation change the status to 'OK' if 0 rows returned, otherwise default status 'Unknown'.



Workspace

None

- fydaccadb (745475582770418)
- fyddevadb (2290997658222571)
- fydmonitoradb (2864876046716896)
- fydpradb (777733219043248)

NEW RELEASE 0.4.0 AVAILABLE AS OF 12-AUG

Key Updates:

- ✓ Simplified GCP setup with Service Principal
- ✓ AWS GovCloud support
- ✓ SAT tagging for deployed jobs
- ✓ Fixes for Azure diagnostics logs check & Azure account API checks
- ✓ Documentation improvements for smoother onboarding
- ✓ Many bug fixes

These enhancements make SAT more robust across GCP, AWS (including GovCloud), and Azure, and help customers adopt it with reduced setup process.

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DEMO TIME

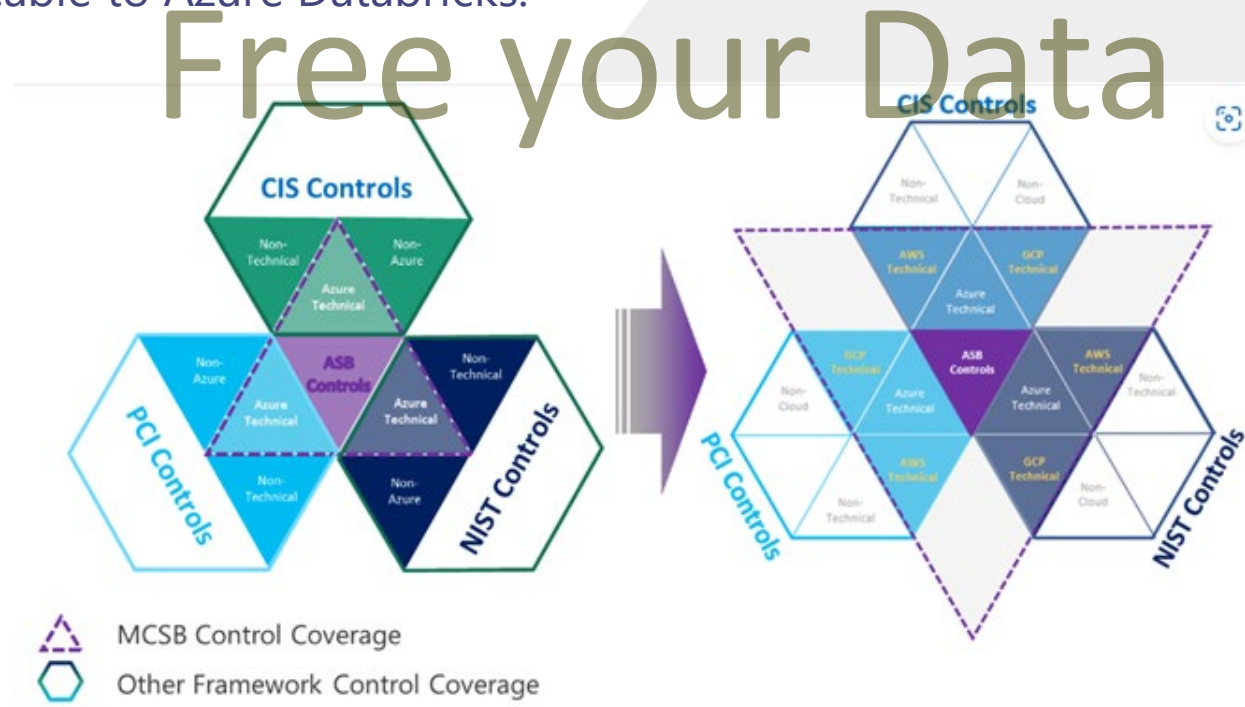
The background of the image is a warm-toned wooden surface. Several wooden rulers are scattered across it. One ruler is positioned horizontally at the bottom, showing markings from 1 to 21. Another ruler is placed diagonally from the top right towards the bottom center, with markings visible from 47 to 60. A third ruler is also visible, partially obscured, running diagonally from the middle right towards the bottom left. A solid blue rectangular overlay covers the left half of the image, serving as a backdrop for the text.

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EXTRA SECURITY MEASUREMENTS

AZURE SECURITY BASELINE FOR AZURE DATABRICKS

This security baseline applies guidance from the [Microsoft cloud security benchmark version 1.0](#) to Azure Databricks. The Microsoft cloud security benchmark provides recommendations on how you can secure your cloud solutions on Azure. The content is grouped by the security controls defined by the Microsoft cloud security benchmark and the related guidance applicable to Azure Databricks.



AZURE SECURITY BASELINE FOR AZURE DATABRICKS

You can monitor this security baseline and its recommendations using Microsoft Defender for Cloud. Azure Policy definitions will be listed in the Regulatory Compliance section of the Microsoft Defender for Cloud portal page.



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The screenshot displays the 'Microsoft Defender for Cloud | Recommendations' interface. The left sidebar contains navigation options: General (Overview, Setup), Recommendations (Attack path analysis, Security alerts, Inventory, Cloud Security Explorer, Workbooks, Community, Diagnose and solve problems), Cloud Security (Security posture, Regulatory compliance, Workload protections, Data and AI security, Firewall Manager, DevOps security), and Management (Environment settings). The main content area shows 'Defender CSPM' with a 'Recommendations by risk' section. A risk-based bar chart indicates 0 Critical, 3 High, 70 Medium, and 9 Low risk items. Below this, a table lists specific recommendations for Azure Databricks Workspaces.

Risk level	Title	Affected resource	Risk
Medium	Azure Databricks Workspaces should be in a virtual network	fydaccadb	
Medium	Azure Databricks Workspaces should be in a virtual network	fyddevadb	
Medium	Azure Databricks Workspaces should be in a virtual network	fydpradb	
Medium	Azure Databricks Workspaces should be in a virtual network	fydcommonadb	
Medium	Azure Databricks Workspaces should be in a virtual network	fydmonitoradb	
Medium	Azure Databricks Workspaces should disable public network access	fydcommonadb	
Medium	Azure Databricks Workspaces should disable public network access	fydmonitoradb	
Medium	Azure Databricks Workspaces should disable public network access	fydaccadb	



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WRAP UP

TAKE AWAYS ARE..

- ✓ SAT is a powerful tool that helps us analyse and monitor Databricks security best-practices measurements.
 - ✓ In the process of setting SAT up, you can configure it to better suit your environment.
- ✓ Provides simple and yet concise reporting solution to evaluate Security
- ✓ It allows us to share information for other Workspace members to take action.
- ✓ Open source and direct contact with the SAT team, as well as Contributions from the community.
 - ✓ Together with Microsoft, we get a full security coverage for Azure and Databricks.

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REFERENCES

- Security Analysis Tool
 - <https://www.databricks.com/blog/announcing-security-analysis-tool-sat>
 - <https://databricks-industry-solutions.github.io/security-analysis-tool/docs/installation/standard/azure/>
 - <https://pypi.org/project/dbl-sat-sdk/>
- Databricks GitHub
 - <https://github.com/databricks>
 - <https://github.com/databricks-industry-solutions/security-analysis-tool>
- Databricks Asset Bundle
 - <https://github.com/databricks/databricks-asset-bundles-dais2023>
 - <https://learn.microsoft.com/en-us/azure/databricks/dev-tools/bundles/>
 - <https://learn.microsoft.com/en-us/azure/databricks/dev-tools/bundles/variables>
- Azure Security Baseline
 - <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/azure-databricks-security-baseline>

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linkedin.com/in/fabiangutierrezardts/



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github.com/FreeYourData



fabian.gutierrez@freeyourdata.nl

Free your Data

THANK YOU!



Rate Saturday Holland



Free your Data



Review to win
With custom Data Saturdays lego stickers!

Rate This session



DOWNLOAD PRESENTATION



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